NPT-8

Infinity Learn by Sri Chaitanya Educational Institutions

NEET PART TEST - 8 (06-01-2025)

SR ELITE (SET-I)

SYLLABUS

Botany : Biomolecules, Cell cycle and cell division, Photosynthesis in higher plants, Respiration in plants, Plant growth and development, Sexual reproduction in flowering plants, Principles of inheritance and Variation, Molecular basis of Inheritance, Biotechnology - Principles & Processes and applications and Microbes in Human welfare

Zoology : Neural Control and Coordination, Chemical coordination and Integration, Human reproduction, Reproductive Health, Principles of inheritance and Variation, Molecular basis of Inheritance, Evolution, Human Health and Diseases, Biotechnological Applications in Medicine, Organisms and Population, Eco system and Biodiversity and conservation.

Physics : Track1: Friction, Wok Power Energy, Collisions, Systems of Particles and Rotational motion, Oscillations, Waves, Dual nature of Radiation and Matter , Atoms, Nuclei, Semiconductors, Ray optics and Optical Instruments and Wave optics: Complete chapter

Track 2: Mechanical Properties of Fluids, Gravitation, Electrical charges and Fields, Electrostatic Potential and Capacitance, Current Electricity, Moving Charges and Magnetism, Magnetism and matter, EMI, AC circuits, EM waves and Units and Measurements.

Expt Skills: Vernier calipers and screw gauge

Chemistry : Track-1: Ionic Equilibrium, Chemical Bonding and Molecular structure, Solutions, Electrochemistry, Chemical Kinetics and Biomolecules

Track-2: Organic chemistry: Alkanes to Diazonium compounds, 13th to 18th group elements, Transition and Inner transition elements and Coordination compounds

Guidelines:

1. **Duration of the Exam is 3 Hours.**
2. **The question paper consists of 180 questions, comprising of 45 questions in each subject.**
3. **Each question carries four marks. One mark will be deducted for every incorrect respons from the total score. Maximum marks are 720.**



BOTANY

1. Bakanae disease is caused by
(1) Bacteria
(2) Virus
(3) Fungus
(4) Hormone deficiency

2. Match Column-I with Column-II

	Column-I		Column-II
A	Allergic pollen	I	*Pinus*
B	Light pollen	II	*Zostera*
C	Sticky pollen	III	*Oryza*(grass)
D	Winged pollen	IV	*Parthenium*
E	Pollen remain viable for months	V	China rose
F	Ribbon like pollen	VI	Rose

	A	B	C	D	E	F
(1)	V	IV	II	III	I	VI
(2)	VI	V	IV	III	II	I
(3)	IV	III	I	II	VI	V
(4)	IV	III	V	I	VI	II

3. Read the following statements
A) Elements present in a sample of earth crust are also present in the sample of living tissues.
B) Saturated fatty acid contain double bond
C) Duration of cell cycle can vary from organism to organism and also from cell type to cell type.
D) In animal cell formation of a new cell wall begins with the formation of a simple precursor called cell plate during cytokinesis
Pick the correct pair
(1) A & B
(2) B & C
(3) C & D
(4) A & C

4. If desired gene is inserted within *Pvu* I and *Pvu* II site on $P^{BR}322$, it would inactivate
(1) tet^r gene and *ori*
(2) amp^r gene and rop gene
(3) amp^r gene and tet^r gene
(4) tet^r gene and amp^r gene

5. Mouse stayed alive and candle continuously burned in Priestly experiment due to
(1) Liberation of oxygen by mint plant during respiration
(2) Liberated of CO_2 by mouse during respiration
(3) Liberation of oxygen by mint plant during photosynthesis
(4) Absorption of O_2 by mint plant during photosynthesis

6. Beta galactoside (Lactose) in *lac* operon is
(1) Enzyme
(2) Inducer
(3) Negative gene regulator
(4) Repressor

7. Read the list of structures given below
A) Golgi complex
B) Endoplasmic reticulum
C) Nucleolus
D) Mitotic/spindle apparatus
Which among them disappear and appear by the end of prophase during mitosis ?

	Disappear	Appear
(1)	A , B, D	C
(2)	C	A, B , D
(3)	A, B, C	D
(4)	D	A, B , C

8. Meristem namely A principally contribute to the elongation of plant along its axis whereas meristem B contribute to increase the girth of plant organs. What are A & B meristems ?

	A	B
(1)	Apical	Intercalary
(2)	Lateral	Apical
(3)	Apical	Lateral
(4)	Intercalary	Apical

9. Chemiosmosis require all except
(1) Thylakoid membrane
(2) Proton gradient
(3) Proton pump
(4) Light harvesting complex

10. What is the vitamin present in a coenzyme NAD ?
(1) Thiamine
(2) Riboflavin
(3) Niacin
(4) Pyridoxine

11. Which one is a product of dedifferentiation in a dicot plant ?
(1) Wood
(2) Mesophyll tissue
(3) Cork cambium
(4) Epidermis

12. How many CO_2, ATP and $NADPH_2$ are needed to make one molecule of glucose through Calvin pathway ?

	CO_2	ATP	NADPH
(1)	6	12	18
(2)	6	18	12
(3)	1	18	12
(4)	1	30	12

13 All the following occur during prophase I of meiosis except
(1) Pairing of homologous chromosomes
(2) Recombinations between homologous chromosomes
(3) Separation of homologous chromosomes and their movement towards two opposite poles
(4) Chromosomes are fully condensed and assembly of meiotic spindle

14. Environmental plasiticity is observerd in
(1) Butter cup
(2) Cotton
(3) Larkspur
(4) Coriander

15. Read the following statements and pick the correct pair.
A) Respiratory pathway comes into picture during breakdown and synthesis of fatty acids & proteins.
B) Ubiquinone receive reducing equivalents *via* $FADH_2$ that is generated during oxidation of succinate in glycolysis.
C) Continued oxidation of Acetyl CoA *via* TCA cycle require continuous replenishment of NAD^+, FAD and oxalo acetic acid.
D) Lactic acid and alcoholic fermentation more energy than aerobic respiration
Choose correct statements
(1) A & B
(2) B & C
(3) C & D
(4) A & C

16. Read the following statements and pick the correct pair
A) *Haem* is a prosthetic group for peroxidase and catalase.
B) Malonate closely resemble succinic dehydrogenase and act as competitive inhibitor of succinic acid
C) Ligases catalyse the removed of groups from substrate by mechanisms other than hydrolysis leaving double bonds
D) Peptide bond (between two amino acids), glycosidic bond (between two sugars) and ester bond (between phosphate and hydroxyl group of sugar) are formed by dehydration
(1) A & C
(2) B & C
(3) C & D
(4) A & D

17. ATP synthesis in glycolysis is by a process namely
(1) Oxidative phosphorylation
(2) Photo phosphorylation
(3) Substrate level phosphorylation
(4) Chemiosmosis

18. Match Column-I with Column-II

	Column-I		Column-II
A	Antibody	I	Drug used to treat certain kinds of cancer
B	Vinblastin	II	Phospholipid present in cell membrane
C	Lecithin	III	Unsaturated fatty acid with twenty carbons
D	Arachidonic acid	IV	Proteins produced by immune system in the presence of antigens

	A	B	C	D
(1)	III	II	I	IV
(2)	IV	II	I	III
(3)	IV	I	II	III
(4)	IV	III	I	III

19. Which one among the following is enzyme mediated process leading to occurrence of recombinations ?
(1) Disjunction
(2) Synapsis
(3) Crossing over
(4) Denaturation in PCR

20. Lenticels help in uptake of oxygen required for respiration in
(1) Soft green stems
(2) Woody stems
(3) Animals
(4) Leaves

21. Read the following statements and pick the correct pair
A) Water deficiency does not affect phosynthesis.
B) Photorespiration is absent in C_4 plants as they have mechanism that increase the concentration of oxygen at RubsiCO site
C) C_4 acid is transported from mesophyll to bundle sheath cells and a C_3 acid is transported from bundle sheath cells to mesophyll cells during C_4 pathway.
D) Calvin pathway is common to all plants namely C_3, C_4 & CAM plants.
(1) A & B
(2) B & C
(3) C & D
(4) A & D

22. The stage in between two meiotic divisions is called
(1) G_1 phase
(2) G_0 phase
(3) Interkinesis
(4) S phase

23. Which is a key end product formed by glycolytic catabolism of glucose in cytoplasm ?
(1) Acetyl CoA
(2) Pyruvic acid
(3) Lactic acid
(4) Fructose 6 phosphate

24. NAD^+ is reduced to $NADH+H^+$ during all these steps except
(1) Glycolysis
(2) Oxidative decarboxylation of pyruvic acid
(3) Fermentation
(4) Kreb's cycle

25. Match Column-I with Column-II

	Column-I		Column-II
A	Lactate dehydrogenase	I	Incomplete oxidation of glucose under anaerobic conditions in yeast
B	Pyruvate dehydrogenase	II	Production of Acetyl CoA & CO_2 & $NADH+H^+$
C	NADH dehydrogenase	III	Oxidation of NADH produced in matrix during ETS
D	Alcohol dehydrogenase	IV	Muscle cells in oxygen deprivation

	A	B	C	D
(1)	IV	III	II	I
(2)	IV	I	II	III
(3)	I	II	III	IV
(4)	IV	II	III	I

26. Read the comparative table and pick the wrong match

	DNA replication	Transcription
(1)	Entire DNA is copied	Only a part of DNA is copied
(2)	Helicase is involved to open the helix by breaking hydrogen bonds	RNA polymerase is involved to unzip the DNA helix by breaking hydrogen bonds
(3)	DNA primers are needed	RNA primers are needed
(4)	Deoxy ribo nucleoside triphosphates act as substrate and energy source	Ribonucleoside triphosphates act as energy source and substrate

27. Read the following statements and pick the correct pair

A) A hybrid genotype always gives an intermediate phenotype

B) An unmodified allele always produce a non functional enzyme leading to recessive triat

C) F_1 Hybrids in Mendel's experiments contain alleles of contrasting traits but express only one of the parental characters

D) In a polygenic trait, the phenotype reflects the contribution of each allele i.e., the effect of each allele is additive.

(1) A and B

(2) B and C

(3) C and D

(4) A and C

28. Match Column-I with Column-II in relation to Mendelian dihybrid cross

	Column-I		Column-II
A	Genotypes that code for recombinant phenotype	I	YyRr
B	Double homozygous recombinant genotypes	II	YYRR, yyrr
C	Double homozygous parental genotypes	III	YYrr, yyRR
D	Double heterozygous genotype	IV	YYrr, Yyrr, yyRR, yyRr

A B C D

(1) I II III IV

(2) II III I IV

(3) III I IV II

(4) IV III II I

29. Infection, blending and centrifugation are three sequential steps in

(1) PCR

(2) Messelson and Stahl experiment

(3) Hershey & Chase experiment

(4) Avery experiment

30. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A): Each restriction endonuclease recognizes a specific palindromic nucleotide sequence in DNA.

Reason (R): The palindrome in DNA is a sequence of base pairs that reads same on both the strands when orientation of reading is kept different.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false and (R) is true.

31. Match Column-I with Column-II with regard to radioactive / heavy elements and their contribution in various discoveries

	Column-I		Column-II
A	Heavy ^{15}N	I	Trace the intermediates of Calvin cycle
B	Radioactive(^{14}C in CO_2	II	Provide unequivocal proof for DNA as genetic material
C	Radioactive ^{32}P and ^{35}S	III	Proof for semi conservative DNA replication
D	Isotope ^{18}O in H_2O	IV	Source of oxygen released in photosynthesis is water

A B C D
(1) I II III IV
(2) II III I IV
(3) III I II IV
(4) IV III II I

32. RNA*i* is blocking *m*RNA with the help of
(1) *t*RNA
(2) *ds*RNA
(3) *ss*DNA
(4) *ds*DNA

33. Read the following statements and choose the correct one
(1) Water is needed for both pollination and fertilization in all terrestrial flowering plants
(2) Nutritive tissue in seeds is endosperm
(3) Female gametophyte is liberated from sporophyte at 7 celled 8 nucleated stage from parent plant
(4) Both meiosis and fertilization is obligatory in production of apomictic seed

34. It is a genus belonging to deuteromycetes. It is employed in elimination of root borne fungal pathogen. What is that genus ?
(1) *Trichoderma*
(2) *Monascus*
(3) *Saccharomyces*
(4) *Alternaria*

35. Match the column – I with column - II

Column – I		Column - II	
(A)	Tassels	(I)	Orchids
(B)	Achenes & Pseudocarp / false fruit	(II)	Mango
(C)	Nucellar embryos	(III)	Corn
(D)	Fruit with many tiny nonendospermic seeds	(IV)	Strawberry

A B C D
(1) III I II IV
(2) III IV II I
(3) IV III II I
(4) I II III IV

36. Viruses could function as
(1) Biofertilizers
(2) Biocontrol agents
(3) Biogas producers
(4) Flocs in STP

37. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).
Assertion (A) : Pollination in *Vallisnaria* occuer on the surface of water
Reason (R) : *Vallisnaria* is a marine free floating hydrophyte
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false and (R) is true.

38. Cloning vector used in genetic engineering need to possess all characters except
(1) Selectable markers helpful in selection of transformants .
(2) One recognition site for one type of restriction enzyme
(3) *Ori* supporting low copy number
(4) Ori supporting high copy number

39. Pick the incorrect match from the following :
(1) GA : Early seed production in conifers
(2) Auxins : Induction of parthenocarpy in tomato
(3) ABA : Increase tolerance in plants to various stresses
(4) Ethylene : Bolting in cabbage & Beet

40. What is the value of respiratory quotient for fats ?
(1) Equal to one
(2) Less than one
(3) zero
(4) More than one

41. Read the statement
" He postulated the presence of adaptor molecule that would on one hand read the code & on other hand would bind to specific aminoacid"
Who is the scientist & adaptor molecule ?
(1) Holley & *t* RNA
(2) Nirenberg & *m* RNA
(3) Crick & *t* RNA
(4) Crick & *m* RNA

42. Blue white selection in genetic engineering is used to / for
(1) Separation of DNA from host cell
(2) Direct gene transfer
(3) Test the presence of insert in a plasmid / heterologous host
(4) Purify and formulate the gene product

43. Aerobic bacteria and fungi are employed in
(1) Fermentation
(2) Biogas plants
(3) Sewage treatment plants
(4) Bread production

44. Which of the following inspect , recognize and cleave the DNA into fragments?
(1) $P^{BR}322$
(2) DNA polymerase
(3) *Eco*RI
(4) *Taq* DNA polymerase

45. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).
Assertion (A): Cytokinins delay senescence in leaves.
Reason (R): Cytokinins cause nutrient mobilization into leaves.
In light of the above statements, choose the correct answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false and (R) is true.

ZOOLOGY

46. Given below are two statements
Statement(S-I): Malignant tumor cells starve the normal cells by competing for vital nutrients
Statement(S-I): Cancer cells actively divide and grow
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

47. The largest known gene is present on.
(1) 9th chromosome
(2) 1st chromosome
(3) Y– chromosome
(4) X– chromosome

48. Which theory, regarding the origin of life, is alternatively also called "Panspermia"?
(1) Special Creation
(2) Spontaneous Generation
(3) Theory of catastrophism
(4) Spore theory

49. Darwin proposed that new species evolve from ancestral forms by
(1) Accumulation of mutations
(2) Struggle for unlimited resources
(3) Inheritance of acquired characters
(4) Gradual accumulation of small variations

50. In which condition, gene frequency remains constant in a population?
(1) Gene flow
(2) Random mating
(3) Mutation
(4) Genetic drift

51. During the growth of any population more individuals acquire peripheral character value at both ends of the distribution curve which lead to the
(1) Disruption
(2) Directional change
(3) Stabilisation
(4) Centripetal selection

52. Match the following.

List-I		List-II	
A)	Gross deformities of genital organs	i)	*Plasmodium*
B)	Dry, scaly lesions on skin	ii)	*Entamoeba*
C)	Stools with excess mucous and blood clots	iii)	*Microsporum*
D)	Chill and high fever recurring every 3 to 4 days	Iv)	*Wuchereria*
		v)	*Ascaris*

(1) A–v, B–ii, C–iii, D–i
(2) A–iv, B–iii, C–ii, D–v
(3) A–iv, B–iii, C–ii, D–i
(4) A–iii, B–iv, C–ii, D–v

53. Which of the given statement is wrong as per the "Theory of Special Creation"?
(1) All living organisms were created as such
(2) Diversity changes as time changes
(3) Earth is about 4000 years old
(4) Diversity was always same

54. Match the columns.

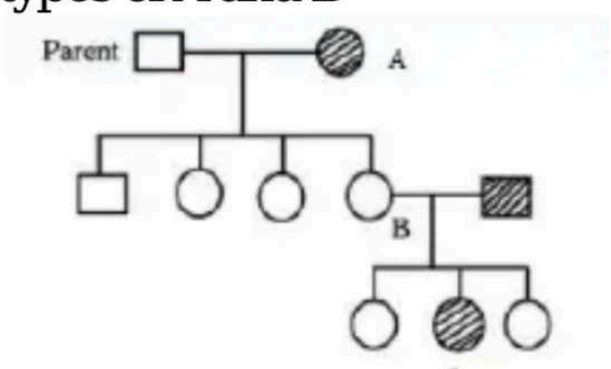
List-I		List-II	
A)	Pre-historic cave art	i)	10,000 years back
B)	Agriculture	ii)	18,000 years back
C)	Neanderthal man	iii)	1.5 mya
D)	*Homoerectus*	iv)	Between 1,00,000-40,000 years back

(1) A–i, B–iii, C – ii, D–iv
(2) A–ii, B–i, C–iv, D–iii
(3) A–ii, B–i, C–iii, D–iv
(4) A–iii, B–i, C–iv, D–ii

55. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion(A): Deficiency of insulin leads to diabetes mellitus
Reason(R): Insulin mainly acts on hepatocytes, adipocytes and enhances cellular glucose uptake and utilisation
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct, but R is not correct

56. Correct order of extinctions through geological time scale is
(1) Pelycosaurs, Therapsids, thecodonts, dinosaurs
(2) Thecodonts, Dinosaurs, Pelycosaurs, Therapsids
(3) Therapsids, Thecodonts, Pelycosaurs, Dinosaurs
(4) Pelycosaurs, Thecodonts, Therapsids, Dinosaurs

57. The given pedigree shows inheritance of autosomal recessive gene. What are the genotypes of A and B



(1) Aa, AA (2) aa, Aa
(3) aa, AA (4) Aa, aa

58. All are characters of Down's syndrome except
(1) Many 'loops' on finger tips
(2) Broad flat face
(3) Congenital heart disease
(4) Small and smooth tongue

59. Find the autosomal dominant trait in the following.
(1) Thalassemia
(2) Phenylketonuria
(3) Sickle-cell anaemia
(4) Myotonic dystrophy

60. In a Hardy-Weinberg population of 1000 individuals, 160 individuals show recessive phenotype, find out the heterozygous dominant individuals in the population
(1) 360 (2) 840
(3) 250 (4) 480

61. Sex of the offspring is determined by the chromosome "W" in case of
(1) Drosophila (2) Grasshopper
(3) Honeybee (4) Pigeon

62. Thalassemia is
(1) A quantitative problem of haemoglobin
(2) A qualitative problem of haemoglobin
(3) Characterised by mental retardation
(4) Characterised by lack of secondary sexual characters

63. Match the following.

List-I		List-II	
A)	Suppress sperm motility	i)	Vasectomy
B)	Makes the cervix hostile to the sperms	ii)	LNG – 20
C)	Increase phagocytosis of sperms	iii)	Lippes loop
D)	Blocks gamete transport	iv)	Multiload 375

(1) A–ii, B–iii, C–i, D–iv
(2) A–iv, B–ii, C–iii, D–i
(3) A–ii, B–iii, C–iv, D–i
(4) A–iv, B–iii, C–ii, D–i

64. Non – specific type of defence is caused by.
(1) T – cells
(2) Antibodies
(3) B – Cells
(4) Neutrophils

65. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion(A): New forms of life have arisen at different times in the history of earth according to paleontological evidences and geological time scale.
Reason(R): A study of fossils showed that life-forms were uniform at different geological time-span and not restricted to any geological time-span.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct, but R is not correct

66. FSH and LH are gonadotropins, in males FSH acts on
(1) Spermatogonia and stimulates the process of spermiation
(2) Leydig cells and stimulates the secretion of androgens
(3) Sertoli cells and stimulates the secretion of some factors which help in spermiogenesis
(4) Leydig cells and stimulates the secretion of inhibin.

67. Given below are two statements
Statement-I: Oral contraceptive pills inhibit ovulation and implantation
Statement-II: Surgical contraceptive methods are highly effective but their reversibility is very poor.
In light of the above statements, choose the appropriate answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

68. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A): Despite occupying about 70 per cent of the earth's surface, the productivity (annual NPP) of oceans are only 55 billion tons per annum.
Reason (R): In oceans, light acts as a limiting factor and amount of nutrients (such as nitrogen) is less.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct, but R is not correct

69. Secondary lymphoid organs provide the sites for interaction of (A) with the (B), then (A) proliferate to become effector cells.
Fill the blanks with correct option:
(1) (A) – Lymphocytes, (B) – antigens
(2) (A) – Memory cells, (B) – lymphocytes
(3) (A) – Auxillary cells, (B) – antibodies
(4) (A) – Antigens, (B) – antibodies

70. In which of the following interactions both partners are affected?
(1) Sea anemone and clown fish
(2) Flamingoes and resident fishes in South American lakes
(3) Fig tree and pollinator wasp
(4) Epiphytic orchid on a mango branch

71. Given below are two statements
Statement-I: Red colourblindness is X-linked recessive disorder
Statement-II: To form red colourblind son, father should be red colourblind.
In light of the above statements, choose the appropriate answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

72. Consider the following statements about ecological pyramids and choose the correct statements:
A) Pyramid of energy can never be inverted, because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step.
B) In parasitic food chain, the pyramid of numbers is upright.
C) The pyramid of biomass of sea is generally inverted because the biomass of fishes far exceeds that of phytoplankton.
(1) Only A (2) Only A and B
(3) Only A and C (4) A, B and C

73. Saheli is a new oral contraceptive for females developed by scientists at CDRI, Lucknow. It prevents
(1) Ovulation
(2) Menstruation
(3) Fertilisation
(4) Implantation

74. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion(A): PCR is now routinely used to detect HIV in suspected AIDS patients
Reason(R): Anti HIV antibodies are detected by PCR.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct, but R is not correct

75. Match the following:

List-I	List-II
A) Correction of a genetic defect	(i) Injecting functional ADA to the patient
B) Pro-insulin	(ii) Delivery of a normal gene into the individual
C) Enzyme replacement therapy	(iii) Needs to be processed to become mature and functional
D) Recombinant therapeutics	(iv) Do not induce unwanted immunological responses

(1) A–iii, B–ii, C–i, D–iv
(2) A–ii, B–iii, C– iv, D–i
(3) A–ii, B–iii, C–i, D–iv
(4) A–iii, B–ii, C–iv, D–i

76. In the ovary of female human, degeneration of a large number of primary follicles occurs during which phase?
(1) From menarche to menopause
(2) From birth to puberty
(3) From foetal stage to birth
(4) From menopause to death

77. Given below are two statements
Statement-I(S-I): At resting state axonal membrane is comparatively more permeable to sodium ions and nearly impermeable to potassium ions
Statement-II(S-II): Ionic gradients across the resting membrane are maintained by the active transport of ions by the $Na^+ – K^+$ pump
In light of the above statements, choose the appropriate answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

78. Study the following statements.
a) Rapid influx of Na^+ followed by the reversal of the polarity at that site is called depolarisation
b) Impulse transmission across the chemical synapse similar to impulse conduction along a single axon
c) Fibres of the tracts are covered with myelin sheath which constitute the outer part of cerebral hemisphere is called white matter
d) The cerebrum wraps around a structure called thalamus which is a major coordinating centre for sensory and motor signaling
Which of the above are incorrect?
(1) b and c
(2) a and d
(3) a and c
(4) b and d

79. Study the following statements.
a) Along with hypothalamus limbic system regulate sexual behavior, expressions of emotional reactions and motivation
b) Mid brain, pons and cerebellum make up the brain stem
c) Hypothalamic hormones reach the posterior pituitary through Portal circulatory system
d) Excess secretion of growth hormone in children results in a disorder called acromegaly
Which of the above are correct?
(1) a and d
(2) b and c
(3) c and d
(4) Only a

80. Given below are two statements
Statement(I): Hormone that stimulates uterine contractions and milk ejection from the mammary gland is prolactin
Statement(II): Hormone which influences metabolism, pigmentation, menstrual cycles and defense capability is melatonin
In light of the above statements, choose the appropriate answer from the options given below
(1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

81. Protein hormone secreted by thyroid gland is
(1) Thymosine (2) Thyroxine
(3) Parathormone (4) Calcitonin

82. Select the correct statements from the following
a) Opioid receptors are present in our CNS and GI tract.
b) Cannabinoid receptors are present principally in the brain
c) Heroin is known for its effects on cardiovascular system of body
d) Crack is a depressant and slows down body functions
(1) b and d
(2) c and d
(3) a and b
(4) b and c

83. Study the following statements
a) Endocrine gland that plays major role in the development of immune system is Thymus
b) Thymosins promote antibody production to provide cell mediated immunity
c) Underproduction of hormones by the adrenal cortex leads to a disease called Cushing syndrome
d) Epinephrine is derived from the amino acid Tyrosine
Which of the above are incorrect
(1) b and d (2) a and c
(3) b and c (4) a and d

84. During oogenesis second polar body is formed.
(1) In tertiary follicle
(2) In Graafian follicle
(3) Just before ovulation
(4) After entry of sperm into the oocyte

85. Which of these are direct ancestors of turtles?
(1) Synapsids (2) Sauropsids
(3) Pelycosaurs (4) Thecodonts

86. Study the following statements
a) Erythropoietin is secreted by juxtaglomerular cells of kidney
b) Secretion of HCI and pepsinogen in the gastric glands is simulated by GIP
c) Secretion of water and bicarbonate ions in the exocrine pancreas is stimulated by secretin
d) CCK stimulates pancreas to secrete bile juice.
Which of the above are correct?
(1) b and d (2) a and c
(3) b and c (4) c and d

87. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion(A): Extinctions across taxa are not random.
Reason(R): Some groups like amphibians appear to be more vulnerable to extinction
In the light of above statements, choose the most appropriate answer from the options given below.
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct, but R is not correct

88. Due to competition which of the following are appeared to be more adversely affected?
(1) Carnivores
(2) Carnivores and herbivores
(3) Herbivores and plants
(4) Plants and carnivores

89. Which of the following cells have surface antibodies?
(1) T_H cells and plasma cells
(2) Plasma cells and memory B-cells
(3) Memory T_C – cells and plasma cells
(4) B – cells and memory B – cells

90. Match the following hosts and ectoparasites.

List-I		List-II	
A)	Hedge plant	i)	Orchid
B)	Dogs	ii)	Lice
C)	Marine fishes	iii)	Cuscuta
D)	Human	iv)	Ticks
		v)	Copepods

(1) A–iv, B–v, C–i, D–ii
(2) A–i, B–iv, C–v, D–ii
(3) A–iii, B–v, C–iv, D–ii
(4) A–iii, B–iv, C–v, D–ii

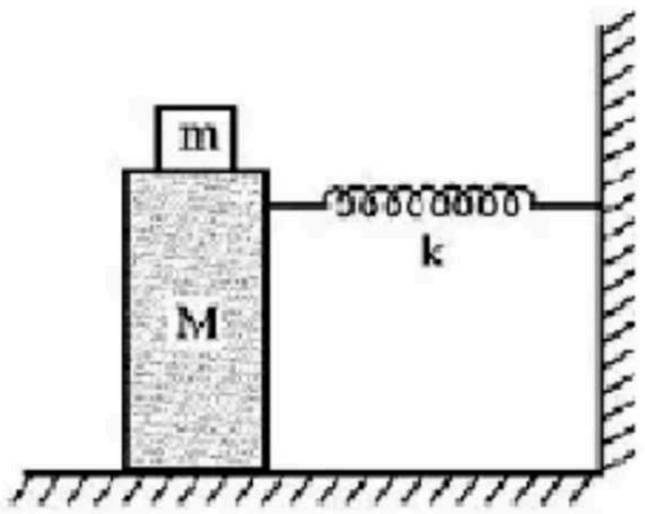
PHYSICS

91. An object of mass m_1 collides with another object of mass m_2, which is at rest. After the elastic collision the objects move with equal speeds in opposite direction. The ratio of the masses $m_2 : m_1$ is
(1) 1 : 2 (2) 3 : 1
(3) 1 : 1 (4) 2 : 1

92. A thin circular ring of mass M and radius is rotating about is axis with an angular speed ω . Two particles having mass m each are now attached at diametrically opposite points. The angular speed of the ring will become.
(1) $\omega\frac{M}{M+m}$ (2) $\omega\frac{M}{M+2m}$
(3) $\omega\frac{M-2m}{M+2m}$ (4) $\omega\frac{M+2m}{M}$

93. A constant power delivering machine has towed a box, which was initially at rest, along a horizontal straight line. The distance moved by the in time 't' is proportional to:
(1) t (2) $t^{2/3}$
(3) $t^{3/2}$ (4) $t^{1/2}$

94. A mass M is attached to a horizontal spring of force constant 'K', oscillates on a frictionless horizontal surface with time period T and amplitude A. When the mass is passing through mean position, another mass m is gently fixed upon it as shown in the figure. The new amplitude of oscillation will be

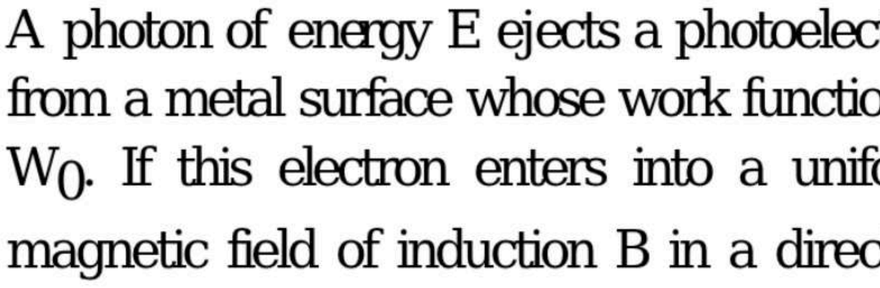


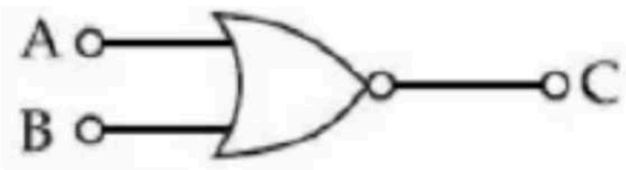
(1) $A\sqrt{\frac{M-m}{M}}$ (2) $A\sqrt{\frac{M}{M+m}}$
(3) $A\sqrt{\frac{M}{M-m}}$ (4) $A\sqrt{\frac{M+m}{M}}$

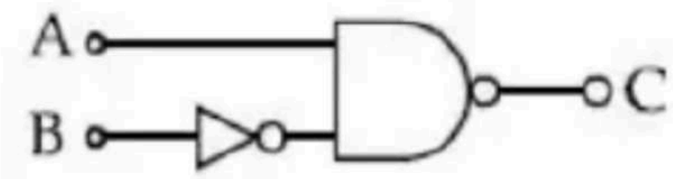
95. A tuning fork A of unknown frequency produces 5 beats/s with a fork of known frequency 340Hz. When fork A is filed, the beat frequency decreases to 2beat/s. What is the frequency of fork A?
(1) 338 Hz (2) 335 Hz
(3) 342 Hz (4) 345 Hz

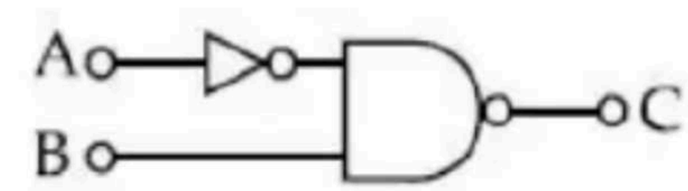
96. The energy that should be added to an electron, to reduce its de-Broglie wavelength from 10^{-10} m to $0.5\times10^{-10} m$, will be
(1) Four times the initial energy
(2) Thrice the initial energy
(3) Equal to the initial energy
(4) Twice the initial energy

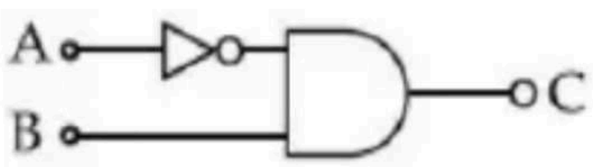
97. A photon of energy E ejects a photoelectron from a metal surface whose work function is W_0. If this electron enters into a uniform magnetic field of induction B in a direction perpendicular to the field and describes a circular path of radius r, then the radius r, is given by (in the usual notation)
(1) $\sqrt{\frac{2m(E-W_0)}{eB}}$ (2) $\sqrt{2m(E-W_0)\,eB}$
(3) $\sqrt{\frac{2e(E-W_0)}{mB}}$ (4) $\frac{\sqrt{2m(E-W_0)}}{eB}$

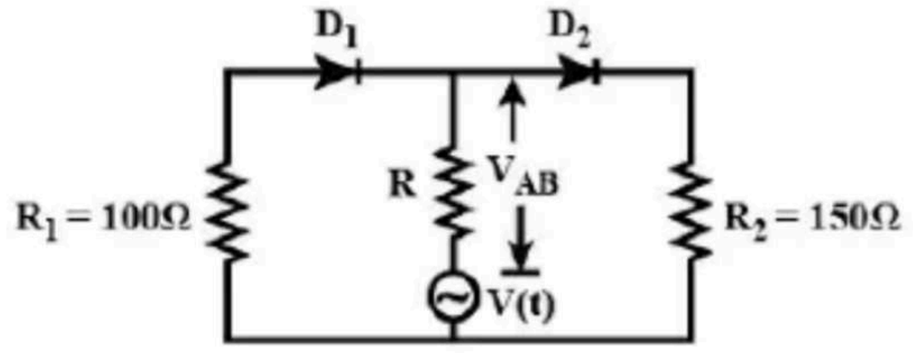
98. 

The logic circuit shown above is equivalent to:
(1) 

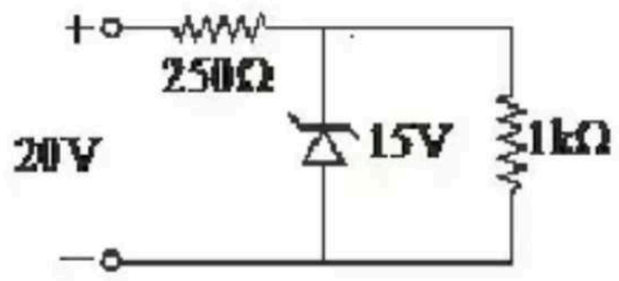
(2) 

(3) 

(4) 

99. In the circuit given below, V(t) is the sinusoidal voltage source, voltage drop $V_{AB}(t)$ across the resistance R is


(1) Is half wave rectified
(2) Is full wave rectified
(3) Has the same peak value in the positive and negative half cycles
(4) Has different peak values during positive and negative half cycle

100. A zener diode, having breakdown voltage equal to 15 V, is used in a voltage regulator circuit shown in figure. The wattage of the diode is


(1) 300 mW (2) 75 mW
(3) 150 mW (4) 225 mW

101. A particle of mass $500 gm$ is moving in a straight line with velocity $v = bx^{5/2}$. The work done by the net force during its displacement from $x = 0\ to\ x = 4m$ is
(Take $b = 0.25 m^{-3/2} s^{-1}$).
(1) 2 J (2) 4 J
(3) 8 J (4) 16 J

102. The ratio of acceleration of the electron in singly ionized helium atom and hydrogen atom (both in ground state) is
(1) 16:1 (2) 8:1
(3) 4:1 (4) 2:1

103. If narrow slit of width 2 mm is illuminated by monochromatic light of wavelength 500 nm, then the distance between the first minima of both side on a screen at a distance of 1 m is
(1) 5 mm (2) 0.5 mm
(3) 1 mm (4) 10 mm

104. A uniform thin bar of mass 6 kg and length 2.4 meter is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon in kgm^2

(1) 0.8 (2) 4.8
(3) 8 (4) 28.8

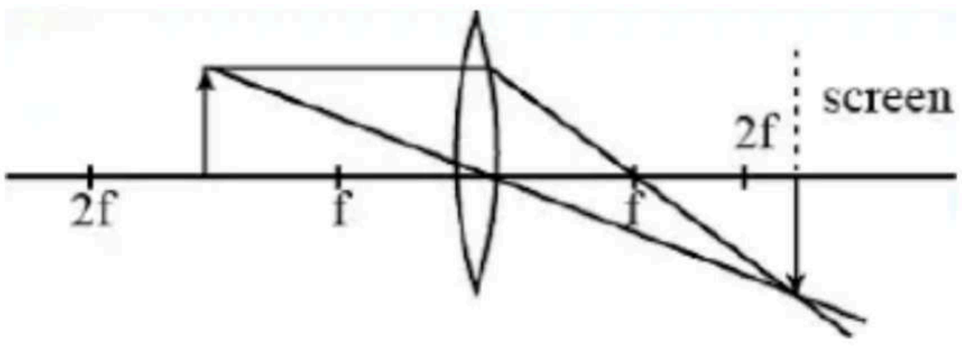
105. The same size images are formed by a convex lens when the object is placed at 20 cm or at 10 cm from the lens. The focal length of convex lens is

(1) 12 cm (2) 40 cm
(3) 18 cm (4) 15 cm

106. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of micro second another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector?

(1) 2 photon of energy 10.2 eV
(2) 2 photon of energy of 1.4 eV
(3) one photon of energy 10.2 eV and an electron of energy 1.4 eV
(4) one photon of energy 10.2 eV and another photon of 1.4 eV

107. Formation of real image using a biconvex lens is shown below.



If the whole set up is immersed in water without disturbing the object and the screen positions, what will one observe on the screen?

(1) Image disappears
(2) Magnified image
(3) Erect real image
(4) No change

108. A block of mass 2 kg rests on a rough inclined plane making an angle of 30^0 with the horizontal. The coefficient of static friction between the block and the plane is 0.7. The frictional force on the block is

(1) 9.8 N
(2) $0.7\times9.8\times\sqrt{3}N$
(3) $9.8\times\sqrt{3}N$
(4) $0.7\times9.8N$

109. In a resonance column lab experiment to measure the velocity of sound, the first resonance is obtained at a length l_1, the second resonance at a length l_2 and third resonance at l_3 then l_3 is

(1) $2l_2-l_1$ (2) l_2-2l_1
(3) l_2-l_1 (4) l_2+l_1

110. If displacement x and velocity v are related as $4v^2=25-x^2$ in a SHM. Then time Period of given SHM is (Consider SI units)

(1) π (2) 2π
(3) 4π (4) 6π

111. The refractive index of a medium is $\sqrt{3}$. If the unpolarised light is incident on it at the polarizing angle of the medium, then angle of refraction is

(1) 60º (2) 45º
(3) 30º (4) 0º

112. A body of mass m is situated in a potential field $U(x)=U_0(1-\cos\alpha x)$ when U_0 and α are constants. The time period of small oscillations is

(1) $2\pi\sqrt{\frac{m}{U_0\alpha}}$
(2) $2\pi\sqrt{\frac{U_0\alpha^2}{m}}$
(3) $\sqrt{\frac{m}{U_0\alpha^2}}$
(4) $2\pi\sqrt{\frac{m}{U_0\alpha^2}}$

113. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion-(A): The four bonding electrons of C, Si and Ge, respectively, lie in 2nd, 3rd and 4th orbit.
Reason-(R): Carbon is insulator while Si and Ge are intrinsic semiconductors even they have same lattice structure.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

114. The nth division of main scale coincides $(n+1)^{th}$ division of vernier scale. Given one main scale division is equal to a units. Find the least count of the vernier.
(1) $\frac{a}{n+1}$ (2) $(n+1)$
(3) $\frac{(n+1)}{a}$ (4) an

115. In an experiment, the values, the values of two resistances were measured to be $R_1 = (5.0 \pm 0.2)\Omega$ and $R_2 = (10.0 \pm 0.1)\Omega$.
Then, their combined resistance in series will be
(1) $(15 \pm 0.1)\Omega$ (2) $(15 \pm 0.2)\Omega$
(3) $(15 \pm 2\%)\Omega$ (4) $(15 \pm 3\%)\Omega$

116. Given below are two statements:
Statement-I: Energy cannot be divided by volume.
Statement-II: Dimensions for energy and volume are different.
In the light of the above statements, choose the correct answer from the options given below.
(1) Statement-I is true, statement-II is true.
(2) Statement-I is true, statement-II is false.
(3) Statement-I is false, statement-II is true.
(4) Statement-I is false, statement-II is false.

117. A particle of mass M is situated at the centre of a spherical shell of same mass and radius a. The gravitational potential at a point situated at $\frac{a}{2}$ distance from the centre, will be
(1) $-\frac{4GM}{a}$ (2) $-\frac{3GM}{a}$
(3) $-\frac{2GM}{a}$ (4) $-\frac{GM}{a}$

118. A spherical body of mass m is moving in a viscous liquid with constant velocity v. Calculate constant velocity of another spherical body of same material and mass 8 m in same liquid.
(1) v (2) 2v
(3) 4v (4) 3v

119. Given below are two statements:
Statement-I: Rise of a liquid in a capillary tube depends on the radius of capillary tube.
Statement-II: Surface tension of a liquid does not depend upon the shaped of vessel in which it is kept.
In the light of the above statements, choose the correct answer from the options given below.
(1) Statement-I is true, statement-II is true.
(2) Statement-I is true, statement-II is false.
(3) Statement-I is false, statement-II is true.
(4) Statement-I is false, statement-II is false.

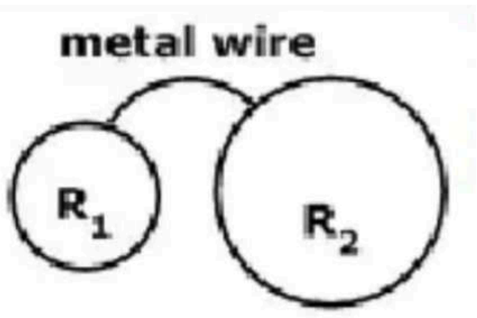
120. For a closed surface through which the net flux is zero, each of the following four statements *could* be true. Which of the statements *must* be true?
(A) There are no charges inside the surface
(B) The net charge inside the surface is zero
(C) The electric field is zero everywhere on the surface
(D) The number of electric field lines entering the surface equals the number leaving the surface
(1) A, C (2) B, D
(3) B, C (4) A, D

121. When a positive charge moves in the direction of the electric field, the electric potential energy of the charge–field system
(1) decreases
(2) increases
(3) remains constant
(4) may increase or decrease

122. Two like charges Q_1 and Q_2 are separated by a distance 'd'. The electric intensity at the location of Q_1 is $\bar{E}_1$ and the electric intensity at the location of Q_2 is $\bar{E}_2$. Which of the following expressions is *zero*?
(1) $Q_1\bar{E}_1 + Q_2\bar{E}_2$ (2) $Q_1\bar{E}_1 - Q_2\bar{E}_2$
(3) $Q_1\bar{E}_2 + Q_2\bar{E}_1$ (4) $Q_1\bar{E}_2 - Q_2\bar{E}_1$

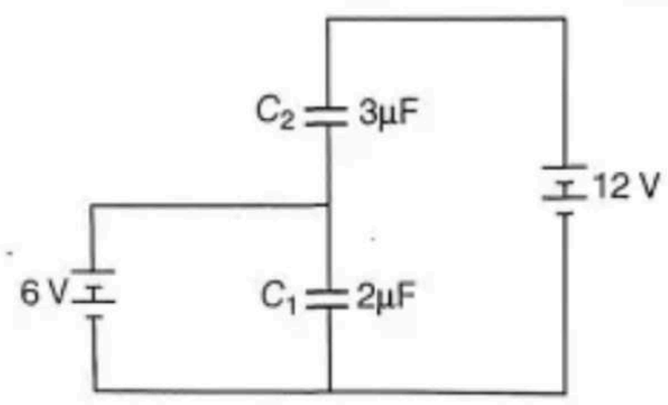
123. 15 joule of work has to be done against an existing electric field to take a charge of 0.01C from A to B. Then, the potential difference $(V_B - V_A)$ is
(1) 1500 V (2) -1500V
(3) 0.15 V (4) None of these

124. Two conducting spheres of radii R_1 and R_2 are connected together by a metal wire. A charge q is given to the system. The ratio of the charge densities σ_1 and σ_2 on their surfaces is



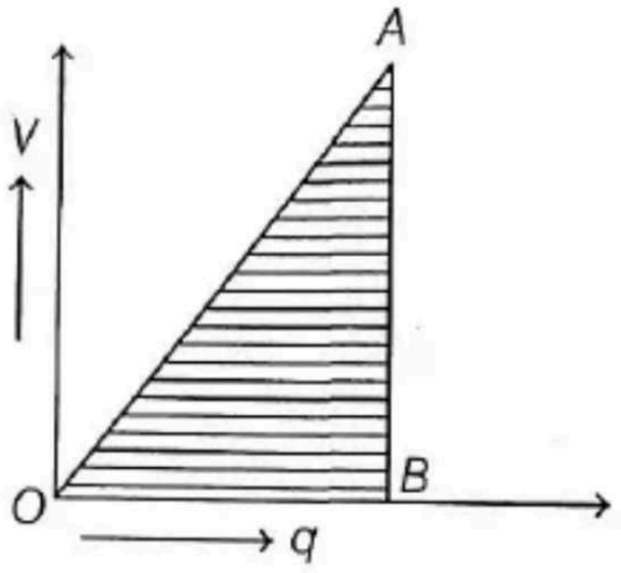
(1) $\frac{R_1^2}{R_2^2}$ (2) $\frac{R_2^2}{R_1^2}$
(3) $\frac{R_1}{R_2}$ (4) $\frac{R_2}{R_1}$

125. Calculate energy stored in capacitance C_1 *and* C_2.



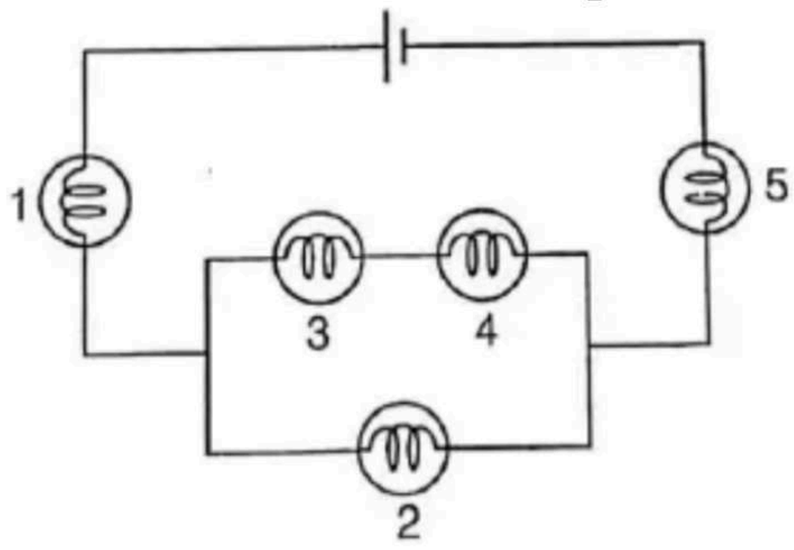
(1) $U_1 = 36\mu J, U_2 = 36\mu J$
(2) $U_1 = 24\mu J, U_2 = 12\mu J$
(3) $U_1 = 36\mu J, U_2 = 54\mu J$
(4) $U_1 = 90\mu J, U_2 = 90\mu J$

126. The charge q on a capacitor varies with voltage as shown in figure. The area of the triangle AOB represents



(1) electric field between the plates
(2) electric flux between the plates
(3) energy density of electric field
(4) energy stored by the capacitor

127. All bulbs in the figure are identical. Which bulb(s) consumes maximum power?

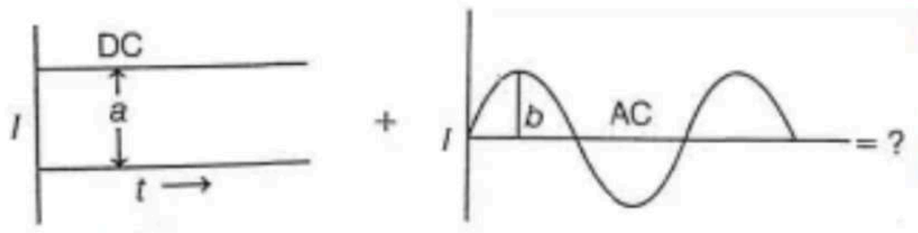


(1) Only 1 (2) Only3 and 4
(3) Only 2 (4) Only 1 and 5

128. Two conductors have the same resistance at 0°C but their temperature coefficients of resistance are α_1 and α_2. The effective temperature coefficient of their series combination is
(1) $\alpha_s = \alpha_1 + \alpha_2$
(2) $\alpha_s = \alpha_1.\alpha_2$
(3) $\alpha_s = \alpha_1 - \alpha_2$
(4) $\alpha_s = \left[\frac{\alpha_1 + \alpha_2}{2}\right]$

129. A proton enters a uniform magnetic field of induction $\vec{B} = x\hat{i} + 0.6\hat{j} + z\hat{k}$ tesla with a velocity $\vec{v} = (2\hat{i} + 3j - 6\hat{k})$ m/s. If the proton continues to move undeviated in the field, then magnitude of the magnetic induction is about
(1) 1.4 T (2) 1.0 T
(3) 14 T (4) 10 T

130. If a direct current of value 'a' ampere is superimposed on an alternative current $I = b\sin\omega t$ flowing through a wire, what is the rms value of the resulting current in the circuit?



(1) $\left[a^2 - \frac{1}{2}b^2\right]^{1/2}$ (2) $\left[a^2 + b^2\right]^{1/2}$
(3) $\left[\frac{a^2}{2} + b^2\right]^{1/2}$ (4) $\left[a^2 + \frac{1}{2}b^2\right]^{1/2}$

131. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion-(A): Self-inductance is called the inertia of electricity.
Reason-(R): The induced emf produced due to self inductance opposes the change of current in the inductor.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) (A) is not correct but (R) is correct.
(2) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(3) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(4) (A) is correct but (R) is not correct.

132. Two choke coils of inductances L_1 and L_2 are connected in series to a battery. Due to changing currents emf's are induced in the coils. The ratio of the induced emf's is
(1) $L_1 : L_2$ (2) $L_2 : L_1$
(3) $L_1^2 : L_2^2$ (4) $L_2^2 : L_1^2$

133. The relative permeability of a material is 0.998. The susceptibility of the material is
(1) 0.002 (2) – 0.002
(3) 1.998 (4) 1.002

134. In an experiment, two bar magnets placed one an another are allowed to oscillate a) with their similar poles on the same side b) with their dissimilar poles on the same side. They are found making 20 and 10 oscillations per minute respectively at the same place. The ratio of their magnetic moments is
(1) 5 : 2 (2) 3 : 5
(3) 1 : 4 (4) 2 : 1

135. A proton, a deuteron and an α particle are moving with same momentum in a uniform transverse magnetic field. The ratio of magnetic forces acting on them and their speed are in the ratio
(1) 1:2:4 and 1:1:2
(2) 1:2:4 and 2:1:1
(3) 4:2:1 and 2:1:1
(4) 2:1:1 and 4:2:1

CHEMISTRY

136. Among the following compounds hydrogen bonding is strongest in
(1) H_2O (2) HF
(3) CH_3CHO (4) NH_3

137. 0.1 F of electricity is passed through aqueous copper sulphate using inert electrodes.
Incorrect statement is
(1) Copper decomposes at anode
(2) pH progressively decreases
(3) 0.05 moles of copper deposits at cathode
(4) Composition of the solution changes

138. Statement-I : For a complex reaction, molecularity has no meaning.
Statement-II : For an irreversible endothermic reaction, enthalpy of reaction is always greater than activation energy.
(1) I is correct, II is incorrect
(2) I and II are correct
(3) I is incorrect, II is correct
(4) I and II are incorrect

139. Glucose and fructose are
(1) Pair of enantiomers
(2) Pair of positional isomers
(3) Pair of functional isomers
(4) Pair of diastereomers

140. Weight of urea that must be dissolved in 500 ml solution whose osmotic pressure is 8.21 atm at 27^0C is
(1) 20 grams (2) 1 gram
(3) 10 grams (4) 2 grams

141. Identify the correct statements
A) Neither pH nor the degree of hydrolysis of CH_3COONH_4 will be affected by dilution.
B) At 298 K, pH of 0.1 M Na_2SO_4 solution is 7
C) Water is neutral at all temperatures
D) As temperature increases, pH of pure water increases
(1) A, B and C only
(2) C and D only
(3) D only
(4) C only

142.

Column-I (Amino acid)	Column –II (Classification)	
I) Valine	A) Neutral	P) Essential
II) Lysine	B) Acidic	Q) Non-essential
III) Tyrosine	C) Basic	
IV) Glutamic acid		

Correct match is
(1) IV-B-P (2) II-C-Q
(3) I-A-P (4) III-C-P

143. 50% of a first order reaction is completed in 1206 days. Time taken for 75% completion of this reaction is
(1) 1809 days
(2) 2412 days
(3) 4824 days
(4) 3618 days

144. Statement-I : Both BrF_5 and $Fe(CO)_5$ have same shape.
Statement-II : Both XeF_4 and $\left[Ni(CN)_4\right]^{2-}$ have same shape and same hybridization.
(1) I is correct, II is incorrect
(2) I and II are correct
(3) I is incorrect, II is correct
(4) I and II are incorrect

145. At certain temperature, vapour pressure of pure water is 45 mm Hg. If 0.6 grams of urea is dissolved in 180 grams of water then the vapour preesure of solution at that temperature would be
(1) 44.955 mm
(2) 41.555 mm
(3) 40.55 mm
(4) 42.55 mm

146. A) Nickel-Cadmium battery
B) Dry cell
C) Lead storage battery
D) Mercury cell
Primary cells in the above list are
(1) A & C (2) B & C
(3) A & D (4) B & D

147. p^{OH} of a mixture containing 50 ml of 0.4 M NH_3 and 50 ml of 0.1 M HCl is
(1) p^{k_b}
(2) $p^{k_b} - \log 3$
(3) $p^{k_b} + \log 3$
(4) $p^{k_b} - 1$

148. Specific conductivity of 0.1 M $NiSO_4$ solution is 'K' Sm^{-1}. Equivalent conductivity of $NiSO_4$ is _____ mho m^2 g eq^{-1}
(1) $\frac{K}{200}$ (2) $\frac{K}{50}$
(3) 10000 K (4) 5000 K

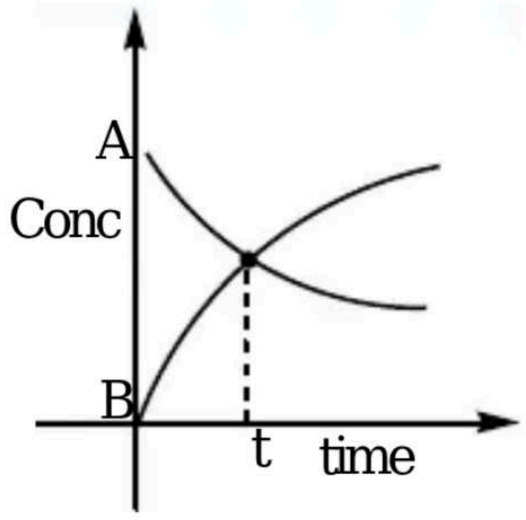
149. All the following species are paramagnetic except.
(1) O_2 (2) B_2
(3) N_2^+ (4) C_2

150. Assertion (A) : NaCl can be precipitated from its saturation solution by passing HCl gas.
Reason (R) : By passing HCl gas through NaCl solution, ionic product of NaCl exceeds its solubility product.
(1) Both A and R are correct but R is not the correct explanation of A
(2) Both A and R are correct R is the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct but R is not correct

151. Which of the following is not present in RNA
(1) Adenine (2) Guanine
(3) Thymine (4) Cytosine

152. Three gases A,B and C are dissolved in water at one bar. Solubility of A, B and C is 0.1 m, 0.2 m and 0.4 m respcectivily. Ratio of $K_H\,(in\ K\ bar)$ for A,B and C is
(1) 4:2:1 (2) 2:3:6
(3) 2:1:4 (4) 6:3:2

153. For the reaction $A \rightarrow 2B$, the following graph is plotted



't' corresponds _____% completion of the reaction.
(1) t_{25} (2) $t_{66.6}$
(3) t_{75} (4) $t_{33.3}$

154. The pair of species which represent Lewis acids is
(1) CO, SO_2 (2) Zn^{2+}, CO_2
(3) CCl_4, PCl_5 (4) $SO_3, CH_2 = CH_2$

155. The reaction in which rate constant has the same units as rate of reaction is
(1) 1 (2) 0
(3) 2 (4) 0.5

156. $CH_4, SF_4, CHCl_3, XeF_4$ and ClF_3. How many molecules possess dipole moment ?
(1) 1 (2) 2
(3) 3 (4) 4

157. A silver electrode in contact with 1M $AgNO_3$ is diluted by ten times. Reduction potential of silver electrode
(1) increases by 0.059 V
(2) increases by 0.0295 V
(3) decreases by 0.059 V
(4) decreases by 0.0295 V

158. Boiling point 0.1m NaCl solution is 100.078^0C. Degree of ionization of NaCl is [k_b of H_2O =0.52 K kg mol^{-1}]
(1) 0.75 (2) 0.5
(3) 0.25 (4) 0.4

159. $C_7H_8 \xrightarrow{3Cl_2/\Delta} A \xrightarrow[dark]{Br_2/FeBr_3} B \xrightarrow{Zn/HCl} C$
Compound 'C' is
(1) Toluene
(2) p-bromotoluene
(3) o-bromotoluene
(4) m-bromotoluene

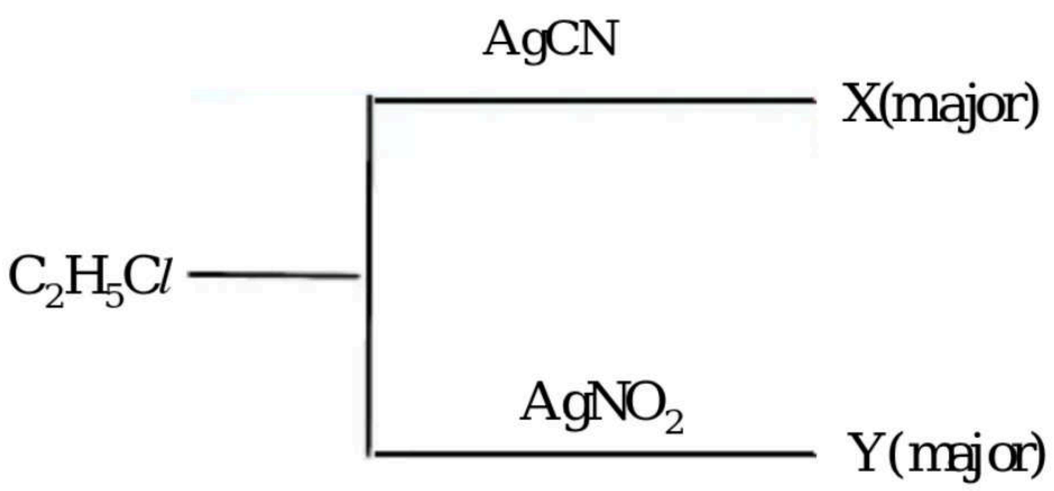
160.

Column-I (13th group element)		Column-II (Property)	
I)	In	A)	Least boiling point
II)	Tl	B)	Least density
III)	B	C)	Least melting point
IV)	Ga	D)	Least ionization enthalpy

Correct match is

	I	II	III	IV
(1)	A	D	B	C
(2)	D	A	B	C
(3)	A	D	C	B
(4)	B	A	C	D

161.



X and Y respectively

(1) CH_3CH_2CN, CH_3CH_2NC

(2) $CH_3CH_2CN, CH_3CH_2NO_2$

(3) $CH_3CH_2NC, CH_3CH_2NO_2$

(4) CH_3CH_2NC, CH_3CH_2ONO

162. $MnO_4^- \xrightarrow[H^+]{I^-} X$. 'X' is

(1) I_2 (2) I_2O_7

(3) IO_4^- (4) I_2O_5

163. Propene

$$\xrightarrow[b)H_2O_2/OH^-]{a)B_2H_6/THF} A \xrightarrow{PCC} B \xrightarrow{LiAlH_4} C.$$

A and C are

(1) 1^0 alcohol, aldehyde

(2) 2^0 alcohol, 2^0 alcohol

(3) 1^0 alcohol, 1^0 alcohol

(4) 1^0 alcohol, carboxylic acid

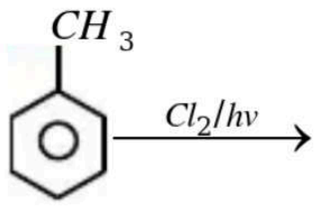
164. Boiling point is least for

(1) N_2 (2) He

(3) O_2 (4) F_2

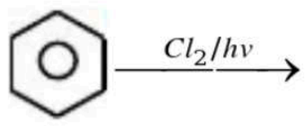
165. Wrong match among the following is

(1)



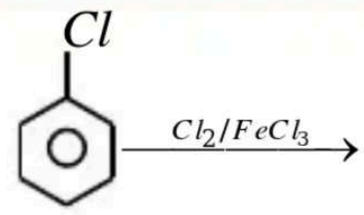
products……free radical substitution

(2)



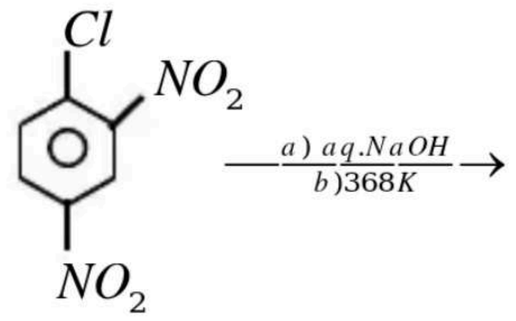
products …….free radical substitution

(3)



Products ….electrophilic substitution

(4)



Products ……..nucleophilic substitution

166. Number of unpaired electrons in $\left[Fe(CN)_6\right]^{4-}$ and $\left[Mn(CN)_6\right]^{3-}$ is respectively.

(1) 0,0 (2) 0,2

(3) 2,2 (4) 2,0

167. Statement-I : Benzyl amine cannot be prepared by Gabriel phthalimide synthesis

Statement-II : 1^0 aryl amines cannot be prepared by Gabriel phthalimide synthesis

(1) I is correct, II is incorrect

(2) I and II are correct

(3) I is incorrect, II is correct

(4) I and II are incorrect

168. Which one of the following can react with steam ?

(1) Pb (2) Ge

(3) Sn (4) Si

169. Benzaldehyde and acetaldehyde can be distinguished using

(1) Tollen's reagent (2) Fehling's reagent

(3) 2,4 – DNP (4) Both 1& 2

170. A) $\left[PtCl_2(NH_3)_2\right]$ shows geometrical isomerism

B) $\left[Co(NH_3)_5ONO\right]SO_4$ can show linkage isomerism

C) $\left[Cr(en)_2Cl_2\right]Cl$ can show both geometrical and optical isomerism

Correct statements are

(1) A & B only (2) A & C only

(3) B and C only (4) A, B and C

171. $CH \equiv CH(I)$

$C_2H_5OH(II)$

$H_2O(III)$

$C_6H_5OH(IV)$.

Correct decreasing order of acidc strength is indicated in

(1) IV, III, II, I (2) IV,II, III, I

(3) I, II, III, IV (4) I, III, II, IV

172. Statement-I : $E^0_{Ce^{+4}/Ce^{+3}}$ is positive

Statement –II : $Ce^{+4}(aq)$ acts like oxidizing agent.

(1) I and II are correct

(2) I is correct, II is incorrect

(3) I is incorrect, II is correct

(4) I and II are incorrect

173. $CH_3-CH_2-COOH \xrightarrow[one\ mole]{Cl_2/RedP} A \xrightarrow{AqKOH} B.$

'B' is

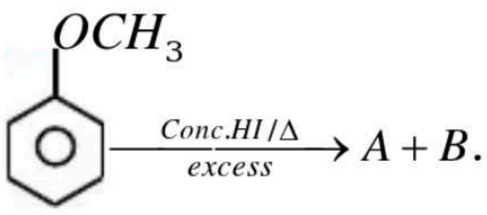
(1) Maleic acid (2) Acetic acid

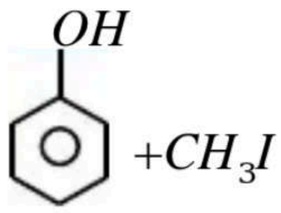
(3) Lactic acid (4) Oxalic acid

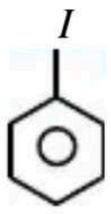
174. Bond angle is highest in

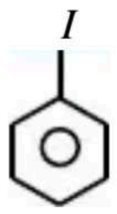
(1) H_2O (2) H_2S

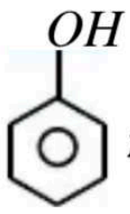
(3) NH_3 (4) PH_3

175.  A& B are

(1)  $+CH_3I$

(2)  ; CH_3I

(3)  ; CH_3OH

(4)  ; CH_3OH

176. Statement-I : Oxygen has least boiling point among 16th group elements.

Statement-II : H_2O is an exothermic product whereas H_2S is an endothermic product.

(1) I is correct, II is incorrect

(2) I and II are correct

(3) I is incorrect, II is correct

(4) I and II are incorrect

177. Most reactive towards S_N2 reaction is

(1) $C_6H_5C(CH_3)_2Br$

(2) $C_6H_5CH_2Br$

(3) $(C_6H_5)_3CBr$

(4) $C_6H_5CH(CH_3)Br$

178. One mole of complex produces four moles of ions in

(1) $[Cr(H_2O)_6]_2(SO_4)_3$

(2) $[Pt(NH_3)_4Cl_2]Cl_2$

(3) $[Co(NH_3)_6]Cl_3$

(4) $[Cr(en)_2Cl_2]Cl$

179. $CH \equiv CH \xrightarrow[2)NaOH+I_2\ 3)H^+]{1)dil.H_2SO_4/Hg^{2+},60^0C} A+B;$

'A' gives positive Tollen's test. Compound 'B' is

(1) CHl_3 (2) CH_3I

(3) CH_2I_2 (4) CI_4

180. The 3d series elements with highest melting point and highest enthalpy of atomization are respectively

(1) Cr, Cr (2) V,V

(3) V,Cr (4) Cr,V